

数学练习 第 07 份 答案与解析

共 75 题 与《练习07》题号一一对应

1. 由 $AA^T = E$ 及 $|A| = 1$,

$$|A - E| = |A - AA^T| = |A||E - A^T| = |E - A| = (-1)^{2n+1}|A - E| = -|A - E|,$$

故 $|A - E| = 0$, 即 $A - E$ 不可逆.

2. $r(A) = 2$, $AX = 0$ 的解为 $k(1, 1 - a, 1)^T$; $B^T = \begin{pmatrix} -1 & -1 & b \\ 1 & 1 & 2 \end{pmatrix}$.

(I) 要求解集真包含, 必须 $\dim N(B^T) \geq 2$, 即 $r(B^T) = 1$, 两行成比例给出 $b = -2$;
再由 $(1, 1 - a, 1)$ 满足 $x_1 + x_2 + 2x_3 = 0$ 得 $1 + (1 - a) + 2 = 0$, $a = 4$. 故 $a = 4$, $b = -2$.

(II) 因 $\dim N(A) = 1$, 有非零公共解当且仅当 $(1, 1 - a, 1)^T \in N(B^T)$, 即

$$-1 - (1 - a) + b = 0, \quad 1 + (1 - a) + 2 = 0 \implies a = 4, b = -2,$$

此时全部非零公共解为 $k(1, -3, 1)^T$ ($k \neq 0$).

3. (I) $A = \begin{pmatrix} a & 1 & b \\ 1 & a & 1 \\ b & 1 & a \end{pmatrix}$. 由 $\text{tr} A = 3a = 1 + 1 + 4 = 6$ 得 $a = 2$;

由 $|A| = 1 \cdot 1 \cdot 4 = 4$ 得 $4 + 2b - 2b^2 = 4$, $b = 0$ 或 1 ; 验证只有 $b = 1$ 时特征值为 $1, 1, 4$.

此时 $A = E + J$, $\lambda = 4$ 的特征向量 $(1, 1, 1)^T$, $\lambda = 1$ 的特征子空间为 $x_1 + x_2 + x_3 = 0$,

$$Q = \left(\frac{(1, -1, 0)^T}{\sqrt{2}}, \frac{(1, 1, -2)^T}{\sqrt{6}}, \frac{(1, 1, 1)^T}{\sqrt{3}} \right).$$

(II) $A^* = 4A^{-1}$, 其特征值为 $\frac{4}{\lambda}$; 故 $A + A^*$ 的特征值均为 $\lambda + \frac{4}{\lambda} = 5$ ($\lambda = 1$ 或 4), 即 $A + A^* = 5E$.
由 $B^2 = 5E$ 且 B 正定得 $B = \sqrt{5}E$.

4. 由递推 $na_n = (\frac{2}{3} - (n-1))a_{n-1}$, 令 $f(x) = \sum a_n x^n$, 则

$$f' - 5 = \frac{2}{3}(f - 3) - xf' \implies (1+x)f' - \frac{2}{3}f = 3.$$

积分因子 $(1+x)^{-2/3}$: $[(1+x)^{-2/3}f]' = 3(1+x)^{-5/3}$,

$$f = -\frac{9}{2} + C(1+x)^{\frac{2}{3}}; \quad f(0) = 3 \implies C = \frac{15}{2}.$$

由二项级数 $(1+x)^{2/3}$ 在 $|x| < 1$ 内收敛知原级数收敛, 且

$$\sum_{n=0}^{\infty} a_n x^n = \frac{15(1+x)^{\frac{2}{3}} - 9}{2}.$$

5. 代入 $\xi = x + ay$, $\eta = x + by$ 得

$$(3a^2 + 4a + 1)u_{\xi\xi} + [2 + 4(a+b) + 6ab]u_{\xi\eta} + (3b^2 + 4b + 1)u_{\eta\eta} = 0.$$

要化为 $u_{\xi\eta} = 0$, 需 $3a^2 + 4a + 1 = 0$ 与 $3b^2 + 4b + 1 = 0$ 且 $a \neq b$, 即

$$\{a, b\} = \left\{ -1, -\frac{1}{3} \right\}$$

(此时交叉项系数 $-\frac{4}{3} \neq 0$).

6. 特征根 $-a \pm \sqrt{a^2 - b^2}$ 均为负, 故 $y, y' \rightarrow 0$ ($x \rightarrow +\infty$). 对方程在 $[0, +\infty)$ 上积分:

$$[y']_0^{+\infty} + 2a[y]_0^{+\infty} + b^2 \int_0^{+\infty} y dx = 0 \implies -1 - 2a + b^2 I = 0,$$

$$\int_0^{+\infty} y(x) dx = \frac{1+2a}{b^2}.$$

7. $EX = 0$, 故用二阶矩: $E(X^2) = 2\theta^2$. 令 $\frac{1}{n} \sum X_i^2 = 2\theta^2$ 得

$$\hat{\theta} = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}, \quad E(\hat{\theta}^2) = \frac{1}{2n} \cdot n \cdot 2\theta^2 = \theta^2.$$

8. 由 $P\{X^2 = Y^2\} = 1$ 知 $\{X = 0\}$ 与 $\{Y = 0\}$ 同时发生, 故

$$P\{X = 0, Y = 0\} = \frac{1}{3}, \quad P\{X = 1, Y = -1\} = P\{X = 1, Y = 1\} = \frac{1}{3},$$

其余为 0. 于是 $EX = \frac{2}{3}$, $EY = 0$, $E(XY) = \frac{1}{3}(-1) + \frac{1}{3}(1) = 0$,

$$\text{Cov}(X, Y) = 0 \implies \rho_{XY} = 0.$$

9. 记 S 为 D 内满足 $y^2 - x^2 > 2$ 的面积, 则 $I = (4\pi - S) - S = 4\pi - 2S$.

两曲线交于 $x = \pm 1$, $y = \pm\sqrt{3}$, 上半部分面积

$$\int_{-1}^1 [\sqrt{4-x^2} - \sqrt{x^2+2}] dx = \frac{2\pi}{3} - 2 \ln \frac{1+\sqrt{3}}{\sqrt{2}},$$

$$\text{故 } S = \frac{4\pi}{3} - 4 \ln \frac{1+\sqrt{3}}{\sqrt{2}},$$

$$I = 4\pi - 2S = \frac{4\pi}{3} + 4 \ln(2 + \sqrt{3}).$$

10. $z = 4 - x - y$, $dS = \sqrt{3} dx dy$, 由对称性 $\iint_{x^2+y^2 \leq 1} (x+y) d\sigma = 0$,

$$I = \sqrt{3} \iint_{x^2+y^2 \leq 1} (4-x-y) d\sigma = 4\sqrt{3}\pi.$$

11. 设该向量在两组基下坐标同为 x , 则 $(\alpha_1, \alpha_2, \alpha_3)x = (\beta_1, \beta_2, \beta_3)x$, 即 $(A - B)x = 0$:

$$A - B = \begin{pmatrix} 0 & -2 & -1 \\ -1 & -3 & -4 \\ 0 & -4 & -2 \end{pmatrix} \implies x = t(5, 1, -2)^T.$$

故该向量为

$$\gamma = 5\alpha_1 + \alpha_2 - 2\alpha_3 = k(4, 5, 2)^T \quad (k \text{ 为任意常数}).$$

12. 列维—林德伯格定理要求独立同分布且方差存在有限. A、B 中同分布但方差未必存在, D 中未必同分布. C 中同服从指数分布, 期望方差都存在. 选 C.

13. 设奇数项之和 $A = \sum u_{2n-1} = 5$, 偶数项之和 B . 由 $\sum (-1)^{n-1} u_n = A - B = 2$ 得 $B = 3$,

$$\sum_{n=1}^{\infty} u_n = A + B = 8.$$

14. L 的方向 $s = (3, -1, 2)$, 过点 $(-2, 2, -1)$. 过 L 且垂直于 π 的平面法向量为

$$s \times n_\pi = (3, -1, 2) \times (2, 3, 3) = (-9, -5, 11),$$

该平面为 $9x + 5y - 11z - 3 = 0$. 故投影直线为

$$\begin{cases} 2x + 3y + 3z - 8 = 0, \\ 9x + 5y - 11z - 3 = 0. \end{cases}$$

15. " (α^T, β^T) 不能由 (A^T, B^T) 的行向量线性表示"恰说明添加这一行后秩增加 1:

$$r \begin{pmatrix} A^T & B^T \\ \alpha^T & \beta^T \end{pmatrix} = r(A^T, B^T) + 1.$$

选 C.

16. $y_1 - y_3, y_2 - y_3$ 是齐次方程的两个线性无关解, 故通解为

$$y = y_3 + C_1(y_1 - y_3) + C_2(y_2 - y_3) = C_1y_1 + C_2y_2 + C_3y_3, \quad C_1 + C_2 + C_3 = 1.$$

选 D.

17. 设切点 (x_0, y_0) , 由隐函数求导得切线在两轴上的截距分别为

$$X = \frac{a}{3x_0 + y_0}, \quad Y = \frac{a}{x_0 + 3y_0},$$

$$S = \frac{XY}{2} = \frac{a^2}{2(3x_0 + y_0)(x_0 + 3y_0)} = \frac{a^2}{2(a + 8x_0y_0)}.$$

由约束 $3(x_0^2 + y_0^2) + 2x_0y_0 = a \geq 8x_0y_0$ 知 $x_0y_0 \leq \frac{a}{8}$ ($x_0 = y_0$ 时取等), 故

$$S_{\min} = \frac{a^2}{2 \cdot 2a} = \frac{a}{4} = \frac{1}{4} \implies a = 1.$$

18. A、B、C 均为标准结论. D 中 X_i 与 S^2 不独立 (S^2 含 X_i), 故 $\frac{X_i^2}{\sigma^2} + \frac{(n-1)S^2}{\sigma^2}$ 不服从 $\chi^2(n)$. 选 D.

19. 对 y 求导得 $f'(x+y) = \frac{(x+y)[(x+y)^2+1]}{f(x+y)}$, 即 $ff' = t^3 + t$,

$$f^2 = \frac{t^4}{2} + t^2 + C; \quad f(1) = \sqrt{2} \implies C = \frac{1}{2},$$

$$\text{故 } f^2 = \frac{(1+x^2)^2}{2}, \quad f(x) = \frac{\sqrt{2}}{2}(1+x^2).$$

20. 在 L 上 $x^2 + y^2 = R^2$, 且 $\oint (e^{x^2} dx - e^{y^2} dy) = 0$ (全微分), 故

$$I = \frac{1}{R^2} \oint_L (-x^2 y dx + xy^2 dy).$$

逆时针时由格林公式 $\oint = \iint_D (x^2 + y^2) d\sigma = \frac{\pi R^4}{2}$, 本题取顺时针,

$$I = -\frac{1}{R^2} \cdot \frac{\pi R^4}{2} = -\frac{\pi R^2}{2}.$$

21. $A^T = -A$ 时 $x^T A x = 0$, 故 $E \pm A$ 均可逆 ($(E+A)x=0 \implies |x|^2=0$).

由分块初等变换:

$$r_1 = n + r(B(E+A)) = n + r(B), \quad r_2 = n + r(AB), \quad r_3 = 2n.$$

而 $r(AB) \leq r(B)$, 故 $r_3 \geq r_1 \geq r_2$. 选 D.

22. 由 $y(0) = y'(0) = 0$ 及方程得 $y''(0) = e^0 - 2y'(0) - y(0) = 1$, 故

$$y(x) = \frac{x^2}{2} + o(x^2) \sim \frac{x^2}{2} \sim \ln \sqrt{1+x^2}.$$

选 D.

23. $V = \frac{4}{3}\pi r^3$, $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 100$, 当 $r = 10$ 时

$$\frac{dr}{dt} = \frac{100}{4\pi \cdot 100} = \frac{1}{4\pi} \text{ cm/s}.$$

24. 实对称矩阵不同特征值的特征向量正交: $\alpha_1 \cdot \alpha_2 = 2 + 3a - 2a = 2 + a = 0 \Rightarrow a = -2$, $\alpha_2 = (1, -2, -4)^T$.
 $|A| = -2$, A^* 的特征值为 $\frac{|A|}{\lambda}$: 对应 $\lambda = 1, 2, -1$ 分别是 $-2, -1, 2$.

故 $(A^* - 2E)X = 0$ 的解空间是 A^* 的特征值 2 (即 A 的特征值 -1) 的特征子空间,
 其特征向量 $\alpha_3 = \alpha_1 \times \alpha_2 \parallel (2, -1, 1)^T$, 通解为

$$X = k(2, -1, 1)^T.$$

25. 列和为零即 $(1, 1, 1)A = 0$, 由对称性 $A(1, 1, 1)^T = 0$, 故 0 是特征值, 结合二重特征值 1 知特征值为 1, 1, 0, A 半正定.

(I) $X^T A X = 0 \iff A X = 0 \iff X \in \text{span}\{(1, 1, 1)^T\}$, 即全部解为

$$X = k(1, 1, 1)^T.$$

(II) $2E - A$ 的特征值为 1, 1, 2, 正定. 取正交矩阵

$$Q = \left(\frac{(1, -1, 0)^T}{\sqrt{2}}, \frac{(1, 1, -2)^T}{\sqrt{6}}, \frac{(1, 1, 1)^T}{\sqrt{3}} \right), \quad Q^T(2E - A)Q = \text{diag}(1, 1, 2),$$

则 $P = \text{diag}(1, 1, \sqrt{2})Q^T$ 满足 $P^T P = 2E - A$, 即 $X^T(2E - A)X = \|PX\|^2$.

26. (I) $S_n \rightarrow +\infty$, 裂项求和:

$$\sum_{n=1}^{\infty} \left(\frac{1}{S_n} - \frac{1}{S_{n+1}} \right) = \frac{1}{S_1} = \frac{1}{u_1}.$$

(II) 当 $n \geq 2$ 时

$$\frac{u_n}{S_n^2} \leq \frac{u_n}{S_{n-1}S_n} = \frac{S_n - S_{n-1}}{S_{n-1}S_n} = \frac{1}{S_{n-1}} - \frac{1}{S_n},$$

右端级数收敛 (裂项), 由比较判别法 $\sum \frac{u_n}{S_n^2}$ 收敛.

27. $f(y)e^x dx + f'(y)e^x dy = d[f(y)e^x]$, 而 O, A 两点处 $y = 0$, $f(0) = 0$, 该部分积分为 0; 又 $\int_L dy = 0$. 故

$$I = -\pi \int_L y dx.$$

补上 $A(-2a, 0)$ 到 O 的 x 轴线段构成逆时针闭曲线, $\oint y dx = -S$, 其中

$$S = \frac{1}{2} \int_0^\pi a^2 (1 - \cos \theta)^2 d\theta = \frac{3\pi a^2}{4},$$

故 $\int_L y dx = -\frac{3\pi a^2}{4}$, $I = \frac{3\pi^2 a^2}{4}$.

28. $f(x) = \frac{1}{\sqrt{\pi}} e^{-(x-1)^2}$, 即 $X \sim N(1, \frac{1}{2})$, 故

$$E(X^2) = DX + (EX)^2 = \frac{1}{2} + 1 = \frac{3}{2}.$$

29. 令 $u = 1 - t: f(x) = \int_{1-x}^1 f(u)du + 1$, 求导得 $f'(x) = f(1-x)$, 再求导 $f''(x) = -f(x)$.

故 $f = C_1 \cos x + C_2 \sin x$; 由 $f(0) = 1$ 得 $C_1 = 1$, 由 $f'(0) = f(1)$ 得 $C_2 = \frac{1 + \sin 1}{\cos 1}$.

$$f(x) = \cos x + \frac{1 + \sin 1}{\cos 1} \sin x.$$

$$30. \quad P\{\max \leq 3\} = P\{X \leq 3\}P\{Y \leq 3\} = \frac{15}{16} \cdot \frac{8}{3e} = \frac{5}{2e}, \quad P\{\max \leq 1\} = \frac{5}{16} \cdot \frac{2}{e} = \frac{5}{8e},$$

$$P\{1 < \max \leq 3\} = \frac{5}{2e} - \frac{5}{8e} = \frac{15}{8e}.$$

31. $P\{a < X < b\} = \Phi(\frac{b}{\sigma}) - \Phi(\frac{a}{\sigma})$. 记 $t = \frac{1}{\sigma}$, 对 t 求导并令其为零:

$$b\varphi(bt) = a\varphi(at) \implies \ln \frac{b}{a} = \frac{(b^2 - a^2)t^2}{2} \implies \sigma^2 = \frac{1}{t^2} = \frac{b^2 - a^2}{2 \ln \frac{b}{a}}.$$

32. 作变换 $u = x + y, v = \frac{x}{x+y}$, 则 $x = uv, y = u(1-v), |J| = u$, 区域变为 $0 \leq u \leq 1, 0 \leq v \leq 1$,

$$\iint_D e^{\frac{x}{x+y}} d\sigma = \int_0^1 \int_0^1 e^v u dv du = \frac{e-1}{2}.$$

33. 在曲线上 $x^2 + y^2 = z$, 故 $d^2 = x^2 + y^2 + z^2 = z + z^2$, 关于 $z \geq 0$ 单调增.

由 $x + y = 4 - z$ 及 $(x + y)^2 \leq 2(x^2 + y^2)$ 得 $(4 - z)^2 \leq 2z$, 即 $z^2 - 10z + 16 \leq 0, 2 \leq z \leq 8$.

$z = 2$ 时 $x = y = 1, d_{\min} = \sqrt{6}$, 点 $(1, 1, 2)$;

$z = 8$ 时 $x = y = -2, d_{\max} = \sqrt{72} = 6\sqrt{2}$, 点 $(-2, -2, 8)$.

$$34. \quad \text{直接计算得 } A^2 = 2A, \text{ 故 } A^n = 2^{n-1}A = 2^{n-1} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

35. 分母 $\sim x^4$; 分子

$$e^{x^2} - e^{2-2\cos x} = e^{2-2\cos x} [e^{x^2-2+2\cos x} - 1] \sim x^2 - 2 + 2\cos x = \frac{x^4}{12} + o(x^4),$$

$$\text{故极限} = \frac{1}{12}.$$

36. 由 $f(tx, ty) = t^2 f(x, y)$ 及欧拉定理 $xf'_x + yf'_y = 2f$. 记 $P = \frac{y}{f}, Q = -\frac{x}{f}$, 则在 D 内

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -\frac{f - xf'_x}{f^2} - \frac{f - yf'_y}{f^2} = -\frac{2f - (xf'_x + yf'_y)}{f^2} = 0.$$

$D = \{y > 0\}$ 是单连通区域且 P, Q 在其中有连续偏导数, 故对任意分段光滑闭曲线积分为零.

37. $\alpha_1 - \alpha_2 = (-1, 1, -1)^T$ 与 $\alpha_1 - \alpha_3 = (0, 1, -2 - a)^T$ ($a \neq -1$ 时线性无关) 都是 $AX = 0$ 的解, 故 $\dim N(A) \geq 2, r(A) \leq 1$, 从而 $r(A) = 0, A^* = O$.

于是 $A^*X = 0$ 的解为全空间, 包含 $AX = 0$ 的解但二者不同解. 选 C.

$$38. \quad \frac{1}{x(a+x^3)} = \frac{1}{a} \left(\frac{1}{x} - \frac{x^2}{a+x^3} \right), \quad \int_1^{+\infty} = \frac{1}{3a} \left[\ln \frac{x^3}{a+x^3} \right]_1^{+\infty} = \frac{\ln(a+1)}{3a}.$$

$$\text{由 } \frac{\ln(a+1)}{3a} = \frac{\ln 2}{3} \text{ 得 } a+1 = 2^a, \text{ 正根为 } a = 1.$$

39. 验证 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} = -\frac{x}{y^2} e^{x/y}$, 是全微分方程. 由

$$u'_x = 1 + e^{x/y} \implies u = x + ye^{x/y} + h(y), \quad u'_y = e^{x/y} \left(1 - \frac{x}{y} \right) + h'(y) = Q \implies h' = 0,$$

故通解为 $x + ye^{\frac{x}{y}} = C$.

40. $f'_y = -2ye^{-x} = 0 \Rightarrow y_0 = 0$; $f'_x = e^{-x}(a - ax - b + y^2) = 0$ 在 $(-1, 0)$ 处给出 $2a - b = 0$.
 又在该驻点 $f''_{xx} = -ae$, $f''_{yy} = -2e$, $f''_{xy} = 0$, 取极大值要求 $f''_{xx} < 0$ 且 $AC - B^2 > 0$, 即 $a > 0$.
 故条件为 $a > 0$, $b = 2a$ (极大值点为 $(-1, 0)$) .

41. 若 (I) 线性相关, 则 $\sum c_i \alpha_i = 0 \Rightarrow \sum c_i A \alpha_i = 0$, (II) 也线性相关; 其逆否命题即 C.
 A、B、D 均有反例 (如 $A = O$) . 选 C .

42. 由 $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 1$ 知 $f(0) = 0$ 且在 0 的去心邻域内 $f(x) > 0 = f(0)$, 故 $f(0)$ 是极小值.
 A、B 中 $\frac{f'(x)}{x}$ 的极限未必存在或不存在; C 中 $f''(0)$ 未必存在. 选 D .

43. (I) 分离变量: $\frac{df}{f^2} = -2x dx \Rightarrow \frac{1}{f} = x^2 + 1$, 即 $f(x) = \frac{1}{1+x^2}$.
 (II) $a_n = \int_{-\infty}^{+\infty} \frac{dx}{(1+x^2)^n} \xrightarrow{x=\tan\theta} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^{2n-2} \theta d\theta$, 故

$$\frac{a_{n+1}}{a_n} = \frac{2n-1}{2n}.$$

$\sum \frac{2n-1}{2n} x^n$ 的收敛半径为 1, 端点处通项不趋于 0, 收敛域 $(-1, 1)$;

$$S(x) = \sum_{n=1}^{\infty} \left(1 - \frac{1}{2n}\right) x^n = \frac{x}{1-x} + \frac{1}{2} \ln(1-x).$$

44. 两曲线交于 $x = 0, 1$, 且 $0 < x < 1$ 时 $x^{3/2} < x$.

$$I_x = \int_0^1 \int_{x^{3/2}}^x y^2 dy dx = \frac{1}{3} \int_0^1 (x^3 - x^{9/2}) dx = \frac{1}{44},$$

$$I_y = \int_0^1 x^2 (x - x^{3/2}) dx = \frac{1}{4} - \frac{2}{9} = \frac{1}{36}.$$

45. 展开行列式得

$$f(x_1, x_2) = ax_1^2 - 2ax_1x_2 + (1-a)x_2^2,$$

其矩阵为 $\begin{pmatrix} a & -a \\ -a & 1-a \end{pmatrix}$, 正定 $\iff a > 0$ 且 $a - 2a^2 > 0 \iff 0 < a < \frac{1}{2}$.

而 A 正定 $\iff 1-a > 0$ 且 $a - 2a^2 > 0 \iff 0 < a < \frac{1}{2}$. 二者等价. 选 C .

46. 在 L 上 $z = 3$ 为常数, $dz = 0$, 且 $\oint 3 dy = 0$, 故 $I = \oint_L x^2 y^3 dx$.
 由格林公式 (逆时针)

$$I = - \iint_{x^2+y^2 \leq 1} 3x^2 y^2 d\sigma = -3 \cdot \frac{\pi}{4} \cdot \frac{1}{6} = -\frac{\pi}{8}.$$

47. 记 $x = \frac{1}{a} \in (0, 1)$,

$$\sum_{k=1}^{\infty} kx^k = \frac{x}{(1-x)^2} = \frac{\frac{1}{a}}{(1-\frac{1}{a})^2} = \frac{a}{(a-1)^2}.$$

48. 相似矩阵行列式相同, 故同时可逆或同时不可逆, B 正确.
 A 显然错; C 错 (特征向量一般不同); D 错 (未必可对角化). 选 B .

49. 弧 $\widehat{OP}$ 与弦 $\overline{OP}$ 所围面积为

$$\int_0^x y(t)dt - \frac{1}{2}xy(x) = x^2.$$

求导得 $y - \frac{1}{2}(y + xy') = 2x$, 即 $xy' - y = -4x$, $\left(\frac{y}{x}\right)' = -\frac{4}{x}$,

$$y = -4x \ln x + Cx; \quad y(1) = 1 \Rightarrow C = 1.$$

故 $y = x - 4x \ln x$ (满足 $y'' = -\frac{4}{x} < 0$, 向上凸) .

50. 驻点 $(\frac{2a}{3}, \frac{2b}{3})$ 处 $f''_{xx} = 2a > 0$, $f''_{yy} = 2b > 0$, $f''_{xy} = 0$, 为极小值点,

$$f\left(\frac{2a}{3}, \frac{2b}{3}\right) = -\frac{4(a^3 + b^3)}{27} = -8 \Rightarrow a^3 + b^3 = 54.$$

椭圆面积 πab 在约束 $a^3 + b^3 = 54$ 下于 $a = b$ 时最大: $a = b = 3$, 最大面积 9π .

51. $EX = 0$, $EX^2 = 1$; $EY = 1$, $DY = \frac{3}{4}$, $EY^2 = \frac{7}{4}$.

$$\text{Cov}(XY, X - Y) = E[X^2Y] - E[XY^2] - 0 = EX^2 \cdot EY = 1,$$

$$D(XY) = EX^2EY^2 = \frac{7}{4}, \quad D(X - Y) = DX + DY = \frac{7}{4},$$

$$\rho = \frac{1}{\sqrt{\frac{7}{4} \cdot \frac{7}{4}}} = \frac{4}{7}.$$

52. 由泰勒公式 $f(x) = x + x^2 + o(x^2)$, 故

$$\lim_{x \rightarrow 0} \frac{f(x) - x}{x^2} = \frac{f''(0)}{2} = 1.$$

53. $F''(x) = f'(x) = 0$ 当且仅当 $x = \mu$ (正态密度的最大值点), 且两侧 F'' 变号,

$$\text{故 } x_0 = \mu, y_0 = F(\mu) = \frac{1}{2}.$$

54. $P\{Z > z\} = P\{X > z\}P\{Y > z\} = e^{-2\lambda z} (z \geq 0),$

$$F_Z(z) = \begin{cases} 1 - e^{-2\lambda z}, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

55. 系数行列式

$$\begin{vmatrix} 1 & a & f(a) \\ 1 & b & f(b) \\ 1 & c & f(c) \end{vmatrix} = (b-a)(c-a) \left[\frac{f(c) - f(a)}{c-a} - \frac{f(b) - f(a)}{b-a} \right].$$

因 $f'' \neq 0$, f 严格凸或严格凹, 两条弦的斜率不相等, 故行列式不为零, 方程组有唯一解.

即三个平面交于一点. 选 A.

56. $\alpha = \tan x(1 - \cos x) \sim \frac{x^3}{2}$, $\beta = \frac{2x^2}{\sqrt{1+x^2} + \sqrt{1-x^2}} \sim x^2$, $\gamma = 1 - \cos(1 - \cos x) \sim \frac{(1 - \cos x)^2}{2} \sim \frac{x^4}{8}$.

阶数由低到高: β (2 阶), α (3 阶), γ (4 阶). 选 D.

57. 锥面上 $z^2 = x^2 + y^2$, $dS = \sqrt{2} dx dy$, 投影区域 $D: x^2 + y^2 \leq 2ax$. 被积函数化为

$$x^2 y + y(x^2 + y^2) + (x^2 + y^2)x,$$

其中含 y 的奇次项因 D 关于 x 轴对称而积分为零, 故

$$I = \sqrt{2} \iint_D x(x^2 + y^2) d\sigma = \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta \int_0^{2a \cos \theta} r^4 dr d\theta = \sqrt{2} \cdot \frac{32a^5}{5} \cdot \frac{5\pi}{16} = 2\sqrt{2}\pi a^5.$$

58. A 可逆时: $(A^*)^{-1} = \frac{A}{|A|} = (A^{-1})^*$, ①正确; $(kA)^* = k^{n-1}A^*$, ②正确;

$(A^*)^T = (A^T)^*$, ③正确; $(A^*)^* = |A|^{n-2}A$, ④正确. 选 D.

59. (I) 幂级数 $\sum (-1)^{n-1} a_n x^n$ 在 $x = e$ 处收敛, 故收敛半径 $R \geq e > 1$, 在 $x = 1$ 处绝对收敛, 即 $\sum a_n$ 绝对收敛.

(II) $\frac{1}{\ln(1+n)} \downarrow 0$, 由莱布尼茨判别法收敛; 而 $\sum \frac{1}{\ln(1+n)} > \sum \frac{1}{n}$ 发散, 故条件收敛.

(III) $\frac{1}{\sqrt{n}-\ln n} \downarrow 0$, 收敛; $\frac{1}{\sqrt{n}-\ln n} \sim \frac{1}{\sqrt{n}}$, $\sum \frac{1}{\sqrt{n}}$ 发散, 故条件收敛.

(IV) $\sqrt[3]{3}-1 \sim \frac{\ln 3}{n} \downarrow 0$, 收敛; 绝对值级数与 $\sum \frac{1}{n}$ 同阶发散, 故条件收敛.

60. $r \parallel a \times b = (-7, -5, -1)$, 设 $r = t(-7, -5, -1)$. 由 $\text{Prj}_c r = \frac{r \cdot c}{|c|} = \frac{-21t}{3} = -7t = 14$ 得 $t = -2$,

$$r = (14, 10, 2).$$

61. $P(\bar{A} | \bar{B}) = 1 - P(A | \bar{B})$, 故条件即 $P(A | B) = P(A | \bar{B})$, 即

$$\frac{P(AB)}{P(B)} = \frac{P(A) - P(AB)}{1 - P(B)} \implies P(AB) = P(A)P(B),$$

即 A 与 B 相互独立. 选 B.

62. 由 $f' = g$, $g' = 1 - f$ 得 $f'' + f = 1$, 且 $f(0) = 1$, $f'(0) = g(0) = 1$, 解得

$$f(x) = 1 + \sin x, \quad g(x) = \cos x.$$

$$I = \int_0^{\frac{\pi}{2}} e^{-x} [\cos x - 1 - \sin x] dx = \left[e^{-x} \sin x + e^{-x} \right]_0^{\frac{\pi}{2}} = 2e^{-\frac{\pi}{2}} - 1.$$

63. 两球面交于 $z = \frac{R}{2}$. 用截面法:

$$I = \int_0^{\frac{R}{2}} \pi z^2 (2Rz - z^2) dz + \int_{\frac{R}{2}}^R \pi z^2 (R^2 - z^2) dz = \frac{\pi R^5}{40} + \frac{47\pi R^5}{480} = \frac{59\pi R^5}{480}.$$

64. $Ax = 0$ 与 $Bx = 0$ 同解 $\iff$ 二者解空间相同 $\iff A, B$ 的行空间相同 $\iff A, B$ 的行向量组等价. 选 B.

65. 在等式两边除以 y 并令 $y \rightarrow 0$ 得 $f'(x) = f(x) - 1$, 即 $f' - f = -1$,

$$f(x) = Ce^x + 1; \quad f(0) = 2 \Rightarrow C = 1,$$

故 $f(x) = e^x + 1$.

66. 记 $Z_i = X_{2i} - X_{2i-1}$, 则 $EZ_i = 0$, $DZ_i = \frac{1}{2}$, $D(\sum_{i=1}^n Z_i) = \frac{n}{2}$.

由中心极限定理, $\frac{\sum Z_i}{\sqrt{n/2}}$ 近似 $N(0, 1)$, 故 $k \cdot \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n/2}}$, 即 $k = \sqrt{2}$.

67. 球面上 $x^2 + y^2 + z^2 = 1$, 且单位外法向为 (x, y, z) , 故 $dydz = x dS$, $dx dy = z dS$:

$$I = \iint_{\Sigma} (x^2 + z^2) dS.$$

Σ 的三条边分别在平面 $z = 0$ 、 $y = 0$ 、 $x = z$ 上, 即 $\Sigma = \{y \geq 0, z \geq 0, x \geq z\}$.

作绕 y 轴的 45° 旋转 $x' = \frac{x+z}{\sqrt{2}}$, $z' = \frac{z-x}{\sqrt{2}}$, 则 $x^2 + z^2 = x'^2 + z'^2 = 1 - y^2$,

区域变为 $\{y \geq 0\} \cap \{-\frac{\pi}{4} \leq \psi \leq 0\}$ (ψ 为绕 y 轴的方位角). 取 $y = \sin \beta$, $dS = \cos \beta d\beta d\psi$:

$$I = \int_{-\frac{\pi}{4}}^0 \int_0^{\frac{\pi}{2}} \cos^2 \beta \cdot \cos \beta d\beta d\psi = \frac{\pi}{4} \cdot \frac{2}{3} = \frac{\pi}{6}.$$

68. 当 $t \rightarrow 0$ 时 $\tan t \sim t$, $\ln(1+t) \sim t$, 故

$$z = \frac{2 \tan t |\ln(1+t)|}{\sqrt{\tan^2 t + \ln^2(1+t)}} \sim \frac{2t|t|}{\sqrt{2}|t|} = \sqrt{2}t,$$

$$\left. \frac{dz}{dt} \right|_{t=0} = \lim_{t \rightarrow 0} \frac{z-0}{t} = \sqrt{2}.$$

选 D.

69. $A = (2-a)E + aJ$, 特征值为 $2+2a$ (对应 $(1,1,1)^T$) 与 $2-a$ (二重, 对应 $x_1 + x_2 + x_3 = 0$).

(I) 取

$$Q = \left(\frac{(1,1,1)^T}{\sqrt{3}}, \frac{(1,-1,0)^T}{\sqrt{2}}, \frac{(1,1,-2)^T}{\sqrt{6}} \right), \quad f = (2+2a)y_1^2 + (2-a)(y_2^2 + y_3^2).$$

(II) $f = \|PX\|^2 = X^T P^T P X$ 且 P 可逆要求 A 正定, 即 $2-a > 0$, 正整数 a 只能取 $a=1$.

此时 A 的特征值为 $4, 1, 1$, 取

$$P = \text{diag}(2, 1, 1) Q^T = \begin{pmatrix} \frac{2}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{2}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \end{pmatrix}, \quad P^T P = A.$$

70. 令 $u = e^f$, 方程化为 $u' - u = x - 1$, 解得 $u = Ce^x - x$; 由 $f(0) = 0$ 得 $C = 1$,

$$f(x) = \ln(e^x - x).$$

(I) 当 $t > 0$ 时 $0 < \ln(e^t - t) < t$, 故 $\{a_n\}$ 单调减且有下界 0 , 极限 L 存在;

由 $L = \ln(e^L - L)$ 得 $L = 0$.

(II) 由 $e^{a_{n+1}} = e^{a_n} - a_n$ 得 $a_n = e^{a_n} - e^{a_{n+1}}$, 故部分和

$$\sum_{n=1}^N a_n = e^{a_1} - e^{a_{N+1}} \rightarrow e - 1,$$

即 $\sum a_n = e - 1$. 又 $a_{n+1} \sim \frac{a_n^2}{2}$, $a_n \rightarrow 0$ 极快, $\sqrt[n]{a_n} \rightarrow 0$,

故幂级数 $\sum a_n x^n$ 的收敛域为 $(-\infty, +\infty)$.

71. 分母 $\sim \frac{x^3}{2}$; 分子中内层 $\int_0^{u^2} \arctan(1+t) dt \sim \frac{\pi}{4} u^2$, 故分子 $\sim \frac{\pi}{12} x^3$,

$$\text{原式} = \frac{\pi/12}{1/2} = \frac{\pi}{6}.$$

72. (I) A 的特征值为 $a+1, a-1, -a$; B 的特征值为 $0, -2, a^2$. 由 $\text{tr}A = \text{tr}B$ 得 $a = a^2 - 2$, $a = 2$ 或 -1 ; 验证只有 $a = -1$ 时两组特征值相同 (均为 $0, -2, 1$). 故 $a = -1$.

(II) 此时 $A = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$, 交换第 2, 3 个坐标即得 B , 故取

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad (\text{正交}), \quad Q^{-1}AQ = Q^T AQ = B.$$

(III) $M = AB = \begin{pmatrix} 1 & -1 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & -1 \end{pmatrix}$ 满足 $M^3 + M^2 + M = O$, 故 $M^4 = M$.

取 $P = -M^2$, 则 $P^2 = M^4 = M = AB$, 即

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

73. 解方程: $(fe^x)' = 1 \Rightarrow f(x) = xe^{-x}$ (由 $f(0) = 0$).

(I) $f' = (1-x)e^{-x}$, $x=1$ 处由正变负, 取极大值 $f(1) = \frac{1}{e}$ (无极小值);
 $f'' = (x-2)e^{-x}$ 在 $x=2$ 处变号, 拐点 $(2, \frac{2}{e^2})$.

(II) 令 $g(x) = f(x) - f(2-x)$ ($0 < x < 1$), 则

$$g'(x) = (1-x)[e^{-x} - e^{x-2}] > 0 \quad (0 < x < 1), \quad g(1) = 0,$$

故 $g(x) < 0$, 即 $f(x_1) < f(2-x_1)$. 又 $f(x_1) = f(x_2)$, 而 $x_2, 2-x_1$ 都在 f 严格减少的区间 $(1, +\infty)$ 上, 由 $f(x_2) < f(2-x_1)$ 得 $x_2 > 2-x_1$, 即 $x_1 + x_2 > 2$.

74. (I) f 为偶函数, $EX = 0$; $DX = EX^2 = \int_0^{+\infty} x^2 e^{-x} dx = 2$.

(II) $X|X|$ 为奇函数, $E(X|X|) = 0$, 故

$$\text{Cov}(X, |X|) = E(X|X|) - EX \cdot E|X| = 0,$$

即 X 与 $|X|$ 不相关.

(III) 不独立: 例如 $P\{|X| < 1, X < -1\} = 0$, 而 $P\{|X| < 1\}P\{X < -1\} > 0$.

75. 在曲线上 $x^2 + y^2 = z$, 故 $d^2 = x^2 + y^2 + z^2 = z + z^2$; 由 $x + y = 1 - z$ 及 $(x+y)^2 \leq 2(x^2 + y^2)$ 得

$$(1-z)^2 \leq 2z \Rightarrow z^2 - 4z + 1 \leq 0 \Rightarrow 2 - \sqrt{3} \leq z \leq 2 + \sqrt{3}.$$

$d^2 = z^2 + z$ 单调增, 故 $d_{\min}^2 = 9 - 5\sqrt{3}$, $d_{\max}^2 = 9 + 5\sqrt{3}$. 引力大小 $F = \frac{1}{d^2}$,

$$F_{\max} = \frac{1}{9 - 5\sqrt{3}} = \frac{9 + 5\sqrt{3}}{6}, \quad F_{\min} = \frac{1}{9 + 5\sqrt{3}} = \frac{9 - 5\sqrt{3}}{6}.$$